
Ancient Heat

7.1 LAUNCH DAY

It's Launch Day. In just a few hours I'll find out if my scientific career is about to move into high gear, or if my future will end up in pieces on the bottom of the Atlantic Ocean.

I start early, waiting for a taxi to take me to Bush House, home of the BBC World Service. Across the UK and Europe other astronomers working on the Herschel and Planck satellites are doing the same. We're all getting media appearances out of the way before heading to wherever we can watch a live feed of the satellites' launch. After my interview, I'll be going to the Rutherford Appleton Lab near Oxford, where the UK Herschel and Planck teams are gathering with visiting dignitaries and members of the media.

I've been working on these two satellites for eight years and they've been a part of the scientific landscape since I completed my PhD in 1991. But success or failure depends on what happens today.

UK media are dominated by BBC television or, for more thoughtful coverage, BBC Radio 4. But if you want a massive audience and worldwide reach you go to the World Service. A top-rated Radio 4 appearance might reach eight million, but the World Service regularly gets audiences of over a hundred million. I try to put thoughts of this vast number of listeners out of my mind as I answer the interviewer's questions. I describe what the satellites will do and what will happen with today's launch. Then it's over and I'm ushered through the splendid art deco interior and into a taxi to Paddington Station for my journey to the Rutherford Lab.

Of the two satellites, Herschel has been around the longest, at least in terms of conceptual development. It is the last of the European Space Agency's cornerstone missions in the Horizon 2000 programme. Originally named FIRST, the Far-Infrared and submillimetre Space Telescope, it's a big expensive mission, costing about a billion Euros. All ESA countries are involved and there are contributions from NASA, Canada and China.

At the Rutherford Lab we've taken over the main lecture theatre. There's an audience of fellow scientists, administrators and the leaders of the funding agencies who paid for the satellites, as well as local politicians and journalists. The partners and children of some of the scientists have also come along so they can find out more about the project that has taken their loved ones away from home, or forced them to work late into the night.

But some people are missing. We're effectively the B-team, left at home to entertain and inform the local media and watch the launch on television. The real bigwigs are seven thousand kilometres away at Kouru, French Guiana, on the coast of South America where ESA launches its rockets. They have ringside seats for the launch and will do their bit for the visiting international media.

The closer we get to the opening of the fifty minute launch window - the period during which the spacecraft have to be launched - the more the tension rises. This isn't just a result of natural excitement. It isn't me worrying about the talk I have to give on the Planck satellite. It's because there's a ghost haunting the party, the ghost of Cluster and the launch of Ariane 501. Back in 1996, the first Ariane 5 rocket was launched carrying the Cluster satellites for the first ESA cornerstone mission. Ariane 5 was a new launch vehicle, the great hope of ESA and Arianespace, its commercial launch company. It could carry larger payloads than previous ESA rockets and was predicted by the engineers to be at least ninety-five percent reliable. I watched the 501 launch on a live video-feed in a lecture theatre not that different from today's. The rocket exploded just seconds after launch, scattering debris over the jungle and into the Atlantic. That explosion was the end of the scientific careers of several friends who had worked on Cluster. Having tied my career to the success or failure of Herschel and Planck, I'm in the same position today as they were before the 501 launch.

I give my talk, and the countdown presses closer to zero. While we have a fifty minute launch window, we aim to launch the instant it opens. Because we're launching two satellites on the same rocket, the launch itself isn't the only hazard. The satellites must get cleanly away from the launch bus before they can make their separate ways to their final destinations. Separation happens thirty minutes after launch, and only at that point can we declare it a success.

There's nothing we can do. When the Vulcain main engine and solid rocket boosters fire, we're entirely dependent on the Arianespace engineers. They have to make sure that hundreds of tonnes of high-explosive rocket fuel explodes in just the right way to get our delicate, high-precision scientific instruments, strapped on top of it all, into orbit without damage.

The final countdown begins.

'Trois, deux, une, toup!'

The Vulcain fires, plumes of steam erupt from either side of the launch pad. Then the solid boosters fire, blasting out distinctive yellow smoke, and the rocket stack is hurled upwards. Across Europe, at JPL, Caltech and online in astronomy groups all over the world, eyes are glued to screens. The roar of the rockets reaches the microphones a few seconds after the rocket clears the pad. By this point, it's already atop a vast column of flame, climbing into a clear blue sky.

Thus begins what must be the longest thirty minutes of my life. We follow the rocket every step of the way. We watch the exhaust plume as it climbs into the sky but, more important, we also follow the telemetry track from the control room, showing how the actual launch is proceeding compared to the prediction for perfect performance.

Soon the camera flicks away from the control room, now filled by our colleagues in Kouru who have seen all they can from the ground, to a view from the rocket itself as the solid fuel boosters are jettisoned - an astounding view from the very edge of space. We ignore the calls from some of the non-astronomers to open the champagne because we know there is still a long way to go.

Next is the separation of the first stage and the ignition of the second. Like every step so far, this goes like clockwork. I begin to think the launch might be going well, but desperately try to suppress these thoughts to avoid the crushing disappointment that will come if some later stage fails. The coverage is now focused on the control room, showing controllers sitting at their desks doing nothing. I am pleased to see they almost look bored.

We reach the final critical phase for Herschel, where it separates from the launch bus. We're treated to a final few frames of video taken by a camera at the back of Herschel showing its clean separation from Planck - the last external view of Planck and its Sylva 5 launch bus that anyone will see. Soon after that, it's time for Planck to cast aside the launcher itself.

Applause erupts. We start breathing again. And outside, at the reception, we hear champagne corks starting to pop.

This afternoon we can celebrate, but tomorrow we start the hard work of commissioning the instruments. It will still be months before real science can begin.

But launch, the most dangerous phase of all, has been a huge success.

7.2 GALAXY EVOLUTION

[Chapter 6](#) looked at the state of galaxies in the local universe. Because they're close, we see them as they are now, at a time that is 13.8 billion years since the Big Bang. We saw that galaxy interactions and mergers are implicated in turning spiral galaxies into ellipticals, in triggering massive bursts of star formation and in fuelling the supermassive black holes at the centres of galaxies, triggering the emergence of active galactic nuclei (AGN). If we look at the universe today we find that about 50% of galaxies are ellipticals, though many of these are dwarf elliptical galaxies that are much smaller than the Milky Way. The fraction of elliptical galaxies is much higher where galaxies are gathered together in galaxy clusters.

If galaxy interactions and mergers turn spiral galaxies into ellipticals, then the ratio of spirals to ellipticals must have been different in the past. The elliptical galaxies we see today represent mergers of two or more galaxies that were previously gas-rich spirals. The merger converted them into an elliptical galaxy through a large burst of star formation and possibly triggered an AGN. If we were to look back in time, we would see the galaxy population changing, with more spirals and fewer ellipticals, and we might also see the star formation rate and the number of AGN increase.

Thanks to the finite speed of light we can make this exact observation. Light takes time to reach us so, in looking at sources that are a very long way away, we see those sources at an earlier stage of the universe. Looking at how the properties of galaxies change with distance, or, equivalently for our purposes here, with increasing redshift, is the study of galaxy evolution. This field has been very active over the last decade, as our observational capabilities have allowed us to identify ever-increasing numbers of galaxies and AGN at greater and greater distances.

Before we were able to study galaxy evolution in detail it was clear that the AGN population changed radically with redshift. Optical spectroscopy of quasars showed that their numbers increase rapidly as we look to higher redshifts. The number of quasars reached a maximum of about thirty times the local value at a redshift of about 2, when the universe was about 3.3 billion years old, roughly a quarter of its age today.

Quasars are much easier to detect and identify at high redshift than galaxies, which is why these results on quasar evolution were arrived at as early as the late 1980s. Similar observations of galaxies require 8 m class telescopes and concerted programmes to obtain spectroscopy of many thousands of galaxies. Studying the shape of galaxies as you look to higher redshift requires instruments capable of resolving very small scale features, since the further away you look the smaller and less distinct the characteristic features of spirals and ellipticals become. Such studies require the resolving power of the Hubble Space Telescope or large ground-based telescopes equipped with instruments that can measure and correct the distorting effects of fluctuations in the atmosphere. All this is required to study the evolution of galaxies in the optical. Studying galaxy evolution in the far-infrared is even more difficult but very necessary because, as we shall see, the optical and far-infrared stories are very different.



Figure 7.1 The original HDF (top) and the HXDF (bottom).

The original HDF (top) is in the optical, the HXDF (bottom) in the optical and near-infrared, with red showing near-infrared light. Each object seen here is a galaxy. Courtesy of NASA.

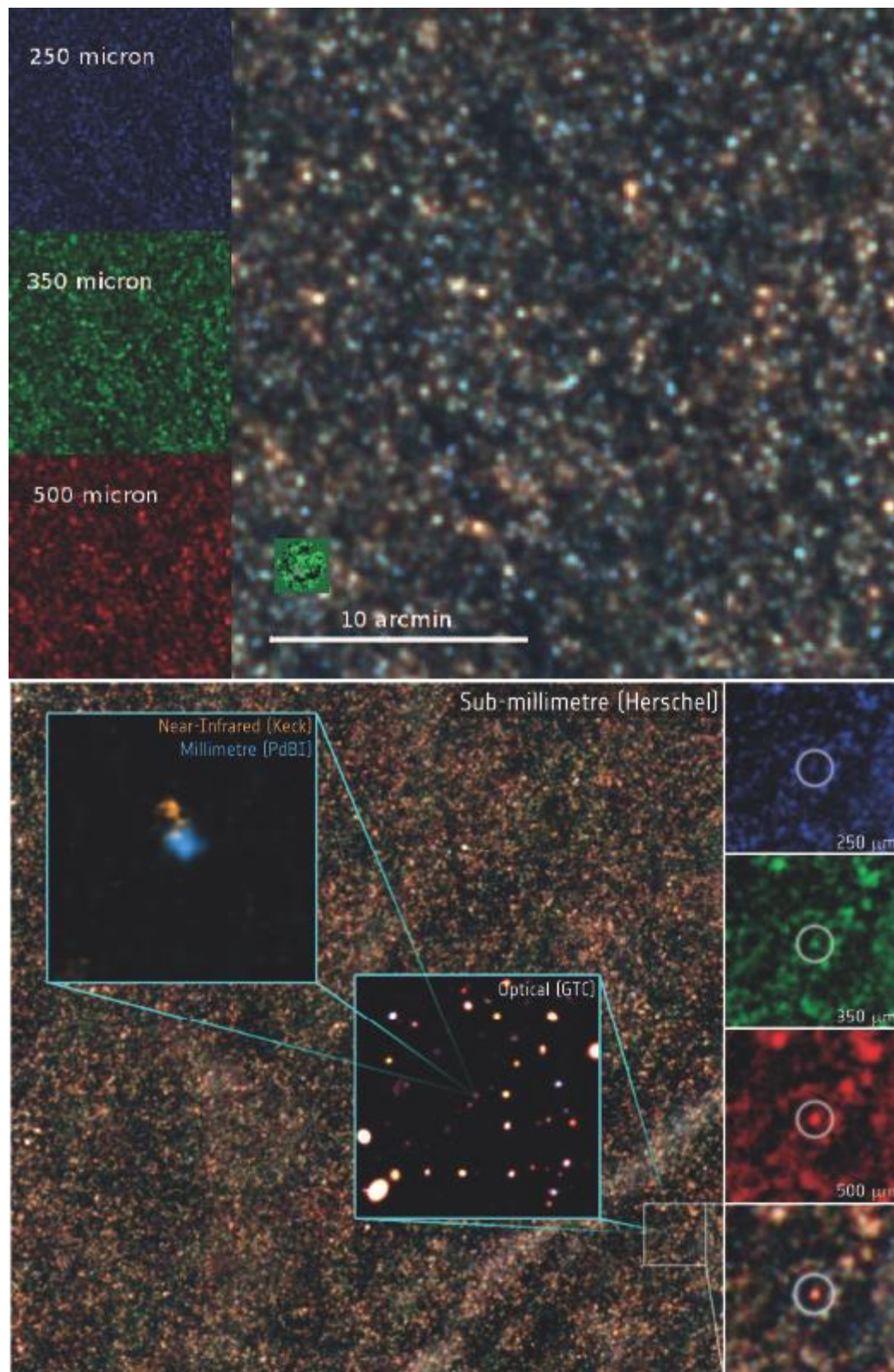


Figure 7.2 Top: HerMES Observations around the HDF; Bottom: The discovery of HFLS3.

Top: Subimages show separate SPIRE channels to left, main image shows a small part of the HerMES survey. Blue is 250 microns, green 350 microns and red 500 microns. Each blob seen here is a galaxy. Inset, to scale, is the 850 micron SCUBA map of the Hubble Deep Field. (Courtesy of ESA.) Bottom: Main image shows the region in which HFLS3 was found, colour scheme as above. At the right are images of HFLS3 at each of these wavelengths and a 'redness image'

showing how it was selected. Insets show region around HFLS3 in the optical (HFLS3 is not detected) and a close-up of dust emission from the Plateau de Bure interferometer and weak near-infrared emission observed by Keck. (Courtesy of ESA.)

7.3 GALAXY EVOLUTION IN THE OPTICAL

To be able to study the evolution of galaxies at different times over the age of the universe, you have to be able to detect them at the distances corresponding to those ages. The further away an object is, the fainter it appears, so the detection of distant galaxies needs a really sensitive instrument. This meant that the study of galaxy evolution started off, and became most fully developed, in the optical, thanks to the sensitivity of 8 to 10 m ground-based telescopes and the Hubble Space Telescope (HST) in orbit.

A defining moment for galaxy evolution studies was the release of the first Hubble Deep Field observations (HDF - see [Figure 7.1](#)) in 1995 [[176](#)]. These observations represented a new paradigm for astronomy in two ways. Firstly, they made use of an unprecedented amount of time on the HST - a total of 10 full days spent with the telescope pointing continuously at a single small patch of sky. This produced the most sensitive optical image that, at the time, had ever been obtained.

Secondly, the HDF observations were made as a community service. Unlike traditional observations, the data did not belong to anybody, so nobody had a head start in analysing the data and producing results. When the data were released in early 1996, everybody could have access to it. There was a feeding frenzy. I took part in it myself, writing a paper from scratch to submission in about two weeks [[32](#)]¹. The principles set by this initial set of observations, and the tremendous scientific success they were to become, led to many similar very sensitive, openly available ‘legacy’ projects, including a series of ever more sensitive Hubble Deep Fields that now stretch into the near-infrared [[83](#)].

A sensitive image is not sufficient for galaxy evolution studies, even when it is the most sensitive image ever obtained. This is because extreme faintness does not immediately indicate extreme distance. A galaxy that is very faint might be very far away, but it is usually more likely that it is actually fairly nearby, and just emitting very little light. A bright thing can appear faint if it is far away, but a faint thing appears faint no matter where it is - and there are many more faint things than bright things. This is a central problem in astronomy - the night sky appears pretty much as it would if it was simply painted on the inside of a sphere, with all objects the same distance away. This was what ancient astronomers thought was going on and remained the general view until Copernicus and Kepler established the heliocentric view of the Solar System. The only objects that the naked eye can tell are moving against this celestial sphere are the planets, given that name because they are ‘planetes’ - Greek for ‘wanderers’. Much of the study of the position of objects on the sky does not require knowledge of their distance, so it is known as spherical astronomy, since it is conducted on the celestial sphere.

But for studies of galaxy evolution we need to know the distance of sources. The usual way for extragalactic astronomers to determine distances is to obtain spectroscopy for a source, from which a redshift can be derived. Then, thanks to the work of Edwin Hubble, who established that the recession velocity of a galaxy is directly proportional to its distance, we can obtain a distance by dividing through by the Hubble constant, now known to be 20.8 km/s per million light years (or 67.8 km/s per Mpc in astronomers’ units) [[129](#)]. Spectroscopic observations take the light from a source and spread it out over a range of wavelengths. This means you need a more sensitive instrument to get spectroscopy for an object than you need to simply get an image of it. For the faint, potentially very distant, objects detected in the HDF there wasn’t, and might never be, a telescope big enough to measure their spectra.

While the direct measurement of optical spectra for HDF galaxies isn’t something that is going to happen soon, there are several ways of getting redshift *estimates* for these galaxies that are accurate enough for the study of galaxy evolution.

These all rely at some level on examination of the observed shape of the spectral energy distribution of galaxies as revealed by their colours. A measurement of these colours was available since the HDF observations were taken in four different optical filters, covering the whole of the optical part of the spectrum. These included a filter at the edge of the ultraviolet, at 300 nm, bluewards of what the eye can perceive, towards the infrared, at a wavelength of 814 nm, at longer wavelengths than the eye can perceive, as well as one in the blue (450 nm) and one in the red (606 nm). The colours given to the objects shown in [Figure 7.1](#) are derived from their different brightnesses in each of these four bands. More recent deep optical surveys such as the Hubble Ultradeep and Extremely Deep fields (HUDF, HXDF) [83] have added extra optical filters and, most recently, have extended these observations into the near-infrared.

The first method used for estimating the redshifts of faint galaxies in deep surveys, and the one still used to search for the most distant galaxies, is known as the dropout method. This relies not so much on the detailed properties of the source being observed, but on the presence of neutral hydrogen gas in the emitting galaxy and in the intergalactic medium along our line of sight to the source.

Neutral hydrogen will be ionised - its single electron is ejected from the atom - if it is hit by a photon with a wavelength less than 91.2 nm. So any ultraviolet light emitted in a galaxy at this or shorter wavelengths will be very strongly absorbed by even a very small amount of neutral hydrogen. The energy emitted by a galaxy as a function of wavelength drops precipitously in the ultraviolet beyond 91.2 nm. Observations that match emitted wavelengths shorter than 91.2 nm will not be able to see the galaxy, while it will be much more detectable at longer wavelengths. This, all other things being equal, gives us a way of estimating the redshift of a galaxy: if a source is detected at long wavelengths but disappears when observed at shorter wavelengths, you can guess that the redshift is such that the wavelength at which it disappears - the wavelength at which it 'drops out' - corresponds to a wavelength of 91.2 nm at the redshift of the source.

But things are a bit more complicated than that, since the light from any distant galaxy has to travel through many millions of light years of intergalactic space to reach us, and that intergalactic space is not entirely empty. Instead, it is filled with clouds of neutral hydrogen. Neutral hydrogen not only absorbs light at wavelengths short enough to cause ionisation, it will also absorb light at wavelengths that match any of its numerous line transitions (see [Chapter 1](#)).

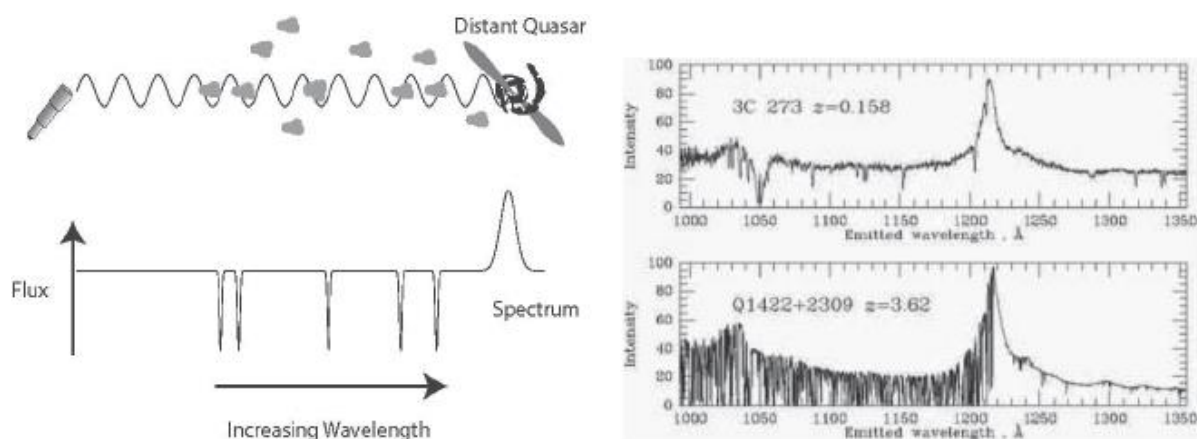


Figure 7.3 The effect of the Lyman- α forest.

Left: A cartoon showing how the Lyman- α forest is produced. Right: Comparison of nearby and high redshift quasar spectra showing the effect of the Lyman- α forest. (After an original by Edward Wright.) Right: Real observations of the effect of the Lyman- α forest, showing a local quasar, 3C273, where there is no forest absorption, compared to a higher redshift object, affected by a plethora of narrow absorption lines. The wavelengths quoted here are those at which the light is emitted, before it has been redshifted. (Figure used with permission from the Astronomy Society of the Pacific.)

One particularly important line transition, which leads to very strong absorption and emission, is at a wavelength of 121.6 nm, and is called the Lyman- α line. Clouds of neutral hydrogen lying between us and a distant source will absorb light from that source at a wavelength matching that of the Lyman- α line at the redshift of the cloud. Since the cloud lies between us and the emitting source, it will be at a lower redshift than the emitter, so that Lyman- α absorption in the cloud will be at a shorter wavelength than that of Lyman- α in the emitter. Conversely, an emitting galaxy will probably have a bright Lyman- α emission line, powered by ultraviolet light coming from young stars and any AGN that might lie at its core.

A cartoon showing the process of Lyman- α absorption by clouds in the intergalactic medium can be found in [Figure 7.3](#). The clouds are very thin, but the Lyman- α transition in hydrogen is so strong that an individual cloud will absorb most of the light of a background source at the wavelength matching that of Lyman- α at its redshift. This leads to a narrow absorption line in the spectrum of the emitting source.

Above a redshift of ~ 2 , these clouds are very numerous. While each cloud only absorbs at a specific wavelength, there are enough of them that they gradually eat away the light of a background source, producing a forest of absorption lines at different wavelengths. In fact these absorption lines are known collectively as the Lyman- α forest. For sources at high redshift they absorb much of the light of a source at wavelengths shorter than that of Lyman- α at the redshift of the source. This means the source drops out of observations at wavelengths shorter than this.

The clouds of hydrogen that produce the forest lines were gradually ionised over time by the ultraviolet light they are absorbing, so there are few, if any, of these clouds lurking in intergalactic space today. This can be seen in [Figure 7.3](#), which shows spectra of the nearby quasar 3C273 and a more distant cousin at a much higher redshift. While the underlying emitted spectrum of the sources are broadly similar, you can see the effect of the Lyman- α forest in the higher redshift source in the way the spectrum is eroded away by many individual narrow Lyman- α absorption lines at wavelengths shorter than the bright emission line (which is Lyman- α emission from the quasars).

The Lyman- α forest and absorption in the emitting galaxy at wavelengths shorter than 91.2 nm mean that any distant galaxy or quasar will have little or no detectable emission at wavelengths shorter than that of Lyman- α at the redshift of emission. In a series of images taken at different wavelengths, such as those taken in the HDF, sources bright at long wavelengths that disappear completely at shorter wavelengths, are likely to be at high redshift - in fact, at a redshift that would put Lyman- α at the wavelength where the source disappears. You can see an example of this from the original HDF in [Figure 7.4](#). This kind of colour-based approach for selecting high redshift objects is quite similar to that used for selecting cool stars in large surveys discussed in [Chapter 4](#).

7.4 REDDER FIELDS, HIGHER REDSHIFTS

Dropout methods were originally applied to optical data from the first Hubble Deep Field and to deep surveys from ground-based optical telescopes. The ground-based surveys are not as sensitive as the HST observations but they can cover much larger areas, so the two types of instrument are very complementary. The original HDF was taken in just four optical bands. Objects that drop out of all except the longest wavelength of these bands would lie at a redshift of about five.

Pushing to higher redshift requires observations in the near-infrared, at wavelengths beyond one micron. Fortunately, the HST is up to the job, firstly with the NICMOS instrument (Near Infrared Camera and Multiobject Spectrometer), and, more recently, with the more sensitive WFC3 instrument (Wide Field Camera 3 - see [Chapter 1](#) for details about the installation and operation of these instruments). WFC3 can observe out to wavelengths of 1.6 microns, so a galaxy visible in this filter that drops out of all shorter wavelengths would potentially be at a redshift of about 10. That corresponds to a time when the universe was 500 million years old, only a few percent of its current age of 13.8 billion years. This is so far

back in time that we really have very little idea of whether galaxies existed, what they might look like, or what kind of stars they might be home to. The discovery of galaxies at such early times would be a major result.

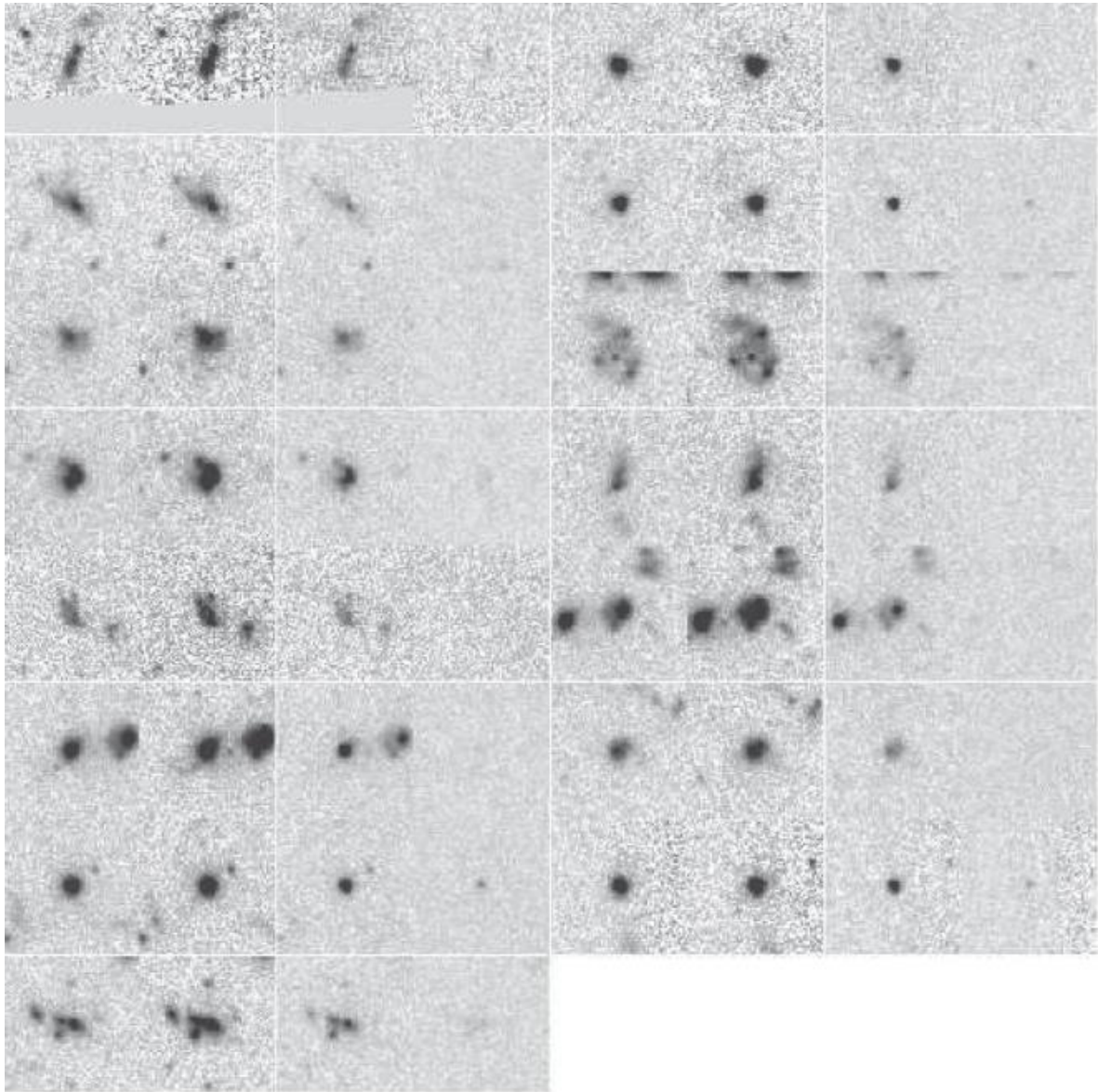


Figure 7.4 Lyman- α dropout galaxies in the HDF-South.

The images show 15 Lyman- α dropout galaxies in the Hubble Deep Field South. The images are in groups of four horizontally, with the rightmost image being at the shortest wavelength, corresponding to the Hubble F300W filter at 300 nm. All these sources are undetected or very faint in this, the shortest wavelength filter, making them candidate dropout galaxies at red-shifts of about three. (Image after one in [33]).

The descendent of the HDF using WFC3 and other updated instruments on the Hubble Space Telescope is called the Hubble Extremely Deep Field, or HXDF. It is shown in [Figure 7.1](#) and is, currently, the deepest optical and near-infrared image ever obtained. This image is the result of the Hubble Space Telescope staring at the same small patch of sky continuously for over three weeks.

When these data were released to the public [83] there was the inevitable feeding frenzy. Astronomers from across the globe searched the catalogs and images for candidate very high redshift galaxies, with dropout selection able to find galaxies not only at redshifts around 10. Also, by looking at sources dropping out at shorter wavelengths, and because of this detected in several longer wavelength bands, galaxies at redshifts from four to eight. could also be found The

advantage of these lower redshift searches is that their sources are detected in several different wavelength images, which adds confidence that they are genuine objects at the indicated redshift. Some tens of galaxies at redshifts out to eight were detected in this way [23].

Armed with a reasonable number of sources at these very great distances, it is possible to investigate the history of galaxy formation, as revealed by optical and near-infrared observations. This is done by working out the amount of energy each source is emitting - termed its luminosity - and then calculating how many sources of each luminosity are found in the volume of the survey. The quantity derived in this way is called the luminosity function, and tracking how it changes with redshift allows us to see how and when galaxies, and the stars inside them, are forming. If most galaxies form at a redshift of, say, six, but a few form at around a redshift of seven, then the luminosity function will be higher at the lower redshift since there will be more galaxies, and more stars inside them, powering them to a higher luminosity.

Figure 7.5, using data from [122], shows how the luminosity function of galaxies selected this way changes as we look from redshifts of four - corresponding to a time when the universe was about 1.5 billion years old - to a redshift of eight, when the universe was less than half that age. Between these redshifts the number of galaxies decreased dramatically, with the number of galaxies of a typical luminosity being a factor of about ten smaller at a redshift of eight than at a redshift of four. The number of high luminosity galaxies decreased by an even larger factor, perhaps by as much as 100, but these sources are so rare at a redshift of eight that hardly any are found in the small area of sky covered by the Hubble deep fields, so the uncertainties on this factor are rather large.

The total amount of star formation underway in the galaxies seen in the Hubble Deep Fields can also be calculated from these luminosity functions.

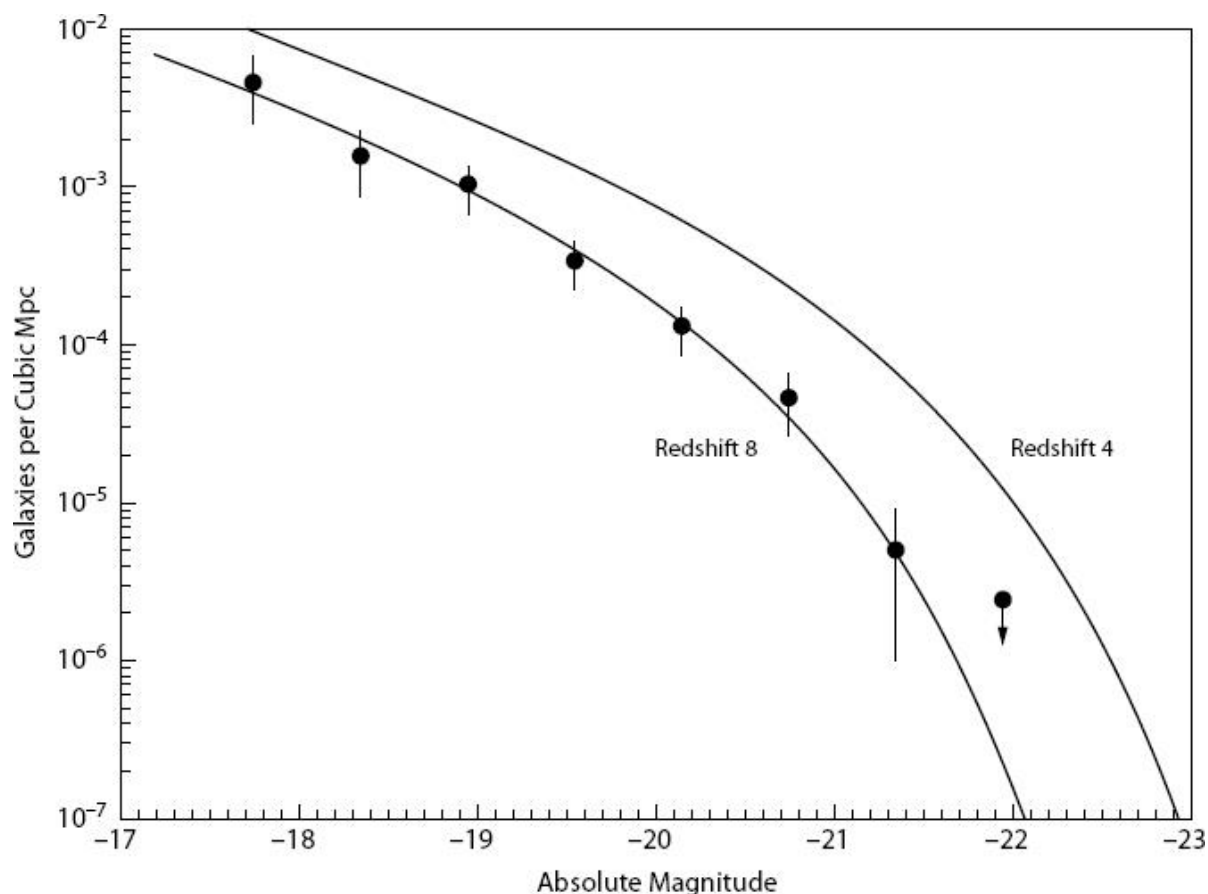


Figure 7.5 Luminosity function evolution for dropout-selected galaxies at a redshift of eight.

This shows how the number of dropout-selected galaxies changes at the highest redshifts. The most luminous galaxies are about a hundred times more common at a redshift of four than they are at a redshift of eight. (Data taken from [122]).

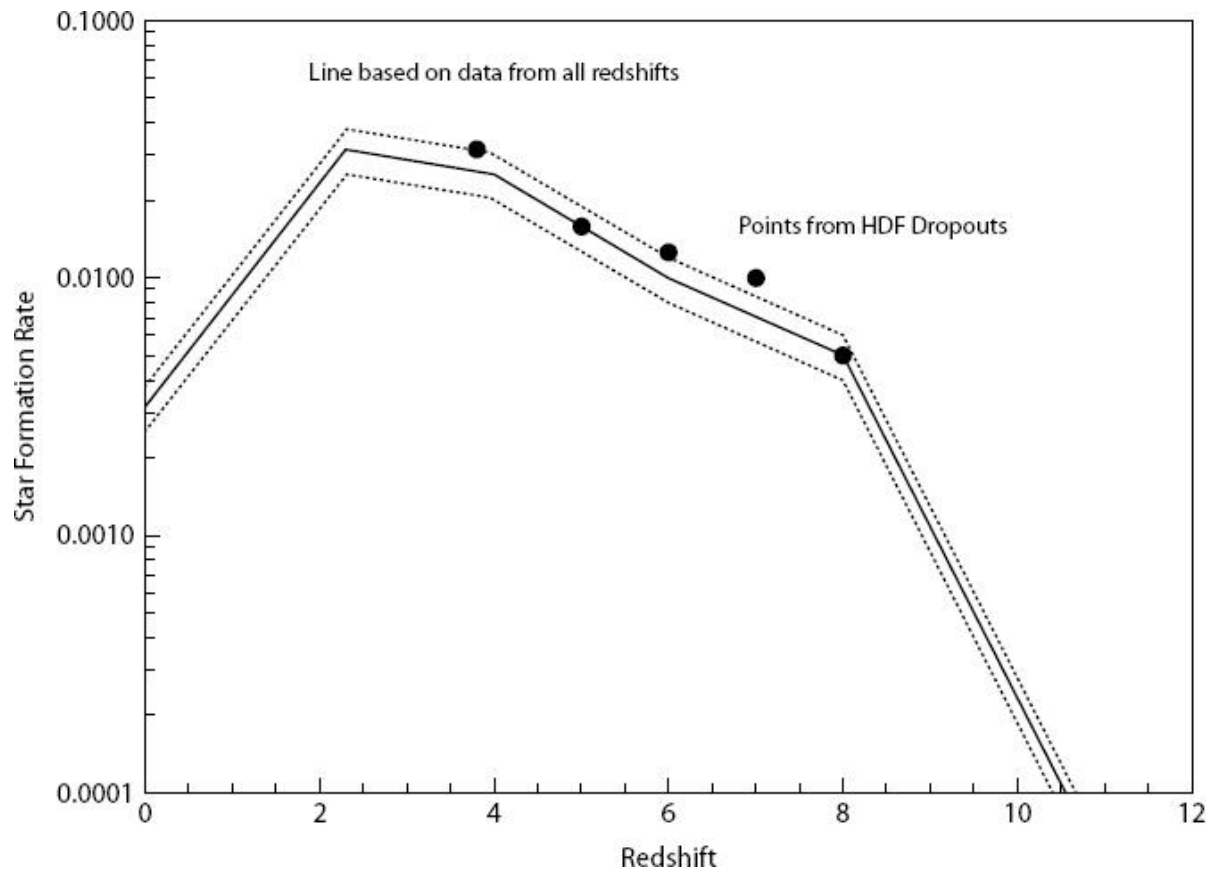


Figure 7.6 The history of star formation in the universe.

The data points at high redshift are taken from [122], while the line shows the results of these data and a compendium of observations at lower redshifts. The star formation rate of the universe, as determined from these observations, peaked at a redshift of about two and then fell. The rate at the present time is about 30 times lower than the rate at its peak.

This is shown in Figure 7.6. You can see that the overall rate of star formation rose steadily from high redshift until it reached a peak at a redshift of about two, corresponding to a time when the universe was about 3.3 billion years old. The star formation rate then steadily declined over the subsequent ten billion years. Today, the star formation rate of the universe is about 30 times less than it was when it reached its peak. The inescapable conclusion from these observations is that the gas that collapses to form new stars is already running out.

7.5 THE MOST DISTANT GALAXIES KNOWN

While the Hubble Deep Fields have provided us with a good number of galaxies at a redshift of eight, they are also capable of finding galaxies at redshifts as great as 10, corresponding to just 500 million years after the Big Bang. Have they detected any such objects?

The first data capable of finding such sources was released in 2009 with the Hubble Ultradeep Field, a predecessor to the deeper HXDF data shown in Figure 7.1. The initial indications were good, with several candidates found that appeared to be detected in just the longest wavelength, 1.6 micron, near-infrared band. But as further work was done, this list, small to begin with, was gradually whittled down to just one candidate that survived all the tests. A paper on this source, potentially at a redshift of 10, was published in the prestigious scientific journal *Nature* [22].

The story doesn't end there, though. More observations have since been collected, producing the even more sensitive HXDF, and the data on the initial redshift 10 candidate has been studied intently by other groups. The problem with a source like this is that it is only detected in a single waveband, and it is so faint that it is very difficult to obtain a spectrum for the source. A good spectrum would provide an unambiguous measurement of the redshift, but without that, there are other possible explanations for its appearance. The source might not be real, for example, but be an unfortunate statistical conspiracy between different sorts of noise that produce the appearance of a source. The deeper HXDF data put paid to that hypothesis - the source is definitely real. However, there are also more subtle alternative possibilities. The source is real, but, rather than lying at a redshift of 10, could it be closer to us? Could its appearance in a single imaging band be due to an unusually bright emission line at an appropriate wavelength in a lower redshift and otherwise very faint galaxy? Could it appear very red not through its high redshift, but as a result of dust obscuration reddening the spectrum of a more nearby object?

These and other ideas have all been checked for this particular source over the last several years, and it now appears that it might not be as distant as it was once thought. The team that originally discovered this object obtained some very sensitive spectroscopic observations with the Hubble Space Telescope that suggest the presence of an emission line from oxygen at the much lower redshift of 2.19 [25]. The brightness of this emission line compared to that of stellar light in the same object is very surprising, but the same study finds other sources with similarly strong emission lines. The unfortunate conclusion is that the first claimed redshift 10 galaxy is probably a much more nearby object.

Further studies, both in the HXDF [123, 53] and elsewhere, have produced other very high redshift candidates, and some of them may well be real. However, the possibility that they are lower redshift sources that are being picked up because of bright emission lines makes it difficult to be absolutely sure about any of them. What is needed is a sensitive near-infrared spectrometer in space, more sensitive than the Hubble Space Telescope can provide. Fortunately such an instrument is planned as part of the James Webb Space Telescope (JWST), due for launch towards the end of this decade. We will say more about this project in [Chapter 9](#).

7.6 THE COSMIC INFRARED BACKGROUND

The picture of high redshift galaxy evolution that has emerged from optical and near-infrared studies over the two decades since the first Hubble Deep Field is one of relatively quiet change. The sort of objects responsible for the star formation history of the universe seen in [Figure 7.6](#) are systems which change slowly, grow stars consistently over a long period of time, and which have little or no excitement in their lives. The kind of fireworks we see in the local universe in objects like Arp220 are not to be found at high redshift in this picture.

But it turns out that this picture is incomplete.

Hints that this is the case, and that far-infrared luminous, starbursting galaxies like Arp220 have a far bigger role in the history of galaxy evolution than suggested by optical studies, first came from IRAS. Studies of the most luminous galaxies detected by this satellite found that their numbers were increasing rapidly with redshift, indicating that they might be much more common at the redshifts where most star formation took place [147]. Of course this is easy to say now, with the benefit of hindsight.

The true significance of dusty starburst galaxies in the history of the universe only became clear in the late 1990s, when the results from two instruments on the Cosmic Background Explorer (COBE) satellite were announced. COBE's main job was to look for small fluctuations in the cosmic microwave background (CMB). This is the dull glow, at a temperature of 2.73 K, left over by the Big Bang. The CMB, the Big Bang, and what it can tell us about the origin of the universe will be the subject of the next chapter.

The two instruments on COBE of interest to us here are the Diffuse Infrared Background Experiment (DIRBE) and the Far-InfraRed Absolute Spectrophotometer (FIRAS). The goal of these two instruments, in the current context, was to search for any other diffuse backgrounds, other than the CMB, that might exist. DIRBE did this by observing the sky with a small, 19 cm, telescope in ten different bands from the near- to far-infrared. FIRAS, in contrast, measured the spectrum of the sky from the far-infrared into the radio, taking an accurate measurement of the amount of energy received at about 40 different wavelengths. It's main job was to determine how closely the CMB matched a black body spectrum², but, once that was determined, the CMB could be removed and any other backgrounds could be examined.

Two completely independent teams analysed the data from these two instruments in search of any background emission that might appear at wavelengths between those of the CMB and those in the optical and near-infrared produced by the light from galaxies like those seen in the Hubble Deep Fields. In 1996, a French team published the surprising result that the FIRAS data revealed a far-infrared background, whose spectrum peaked at a wavelength of about 200 microns [137]. A lot of careful analysis was needed to remove other contaminating sources from this background, and the results remained somewhat noisy, so the paper only described this detection as 'tentative'. Confirmation, though, of the existence of a cosmic infrared background (CIB) arrived two years later with the results of the DIRBE instrument [58]. The two data sets matched perfectly. Even more surprising was the amount of energy in the CIB. When this is compared to the amount of energy emitted by all the distant optical and near-infrared galaxies found in deep surveys such as the Hubble Deep Fields, which can be thought of as the cosmic optical background (COB), the two are found to be almost identical. This means that the sources that power the CIB have, over the history of the universe, generated as much energy as all the stars we can see in all the galaxies observed by our optical telescopes. This can be seen diagrammatically in Figure 7.7, which also shows that both of these backgrounds still contain much less energy than the CMB - not that surprising since the CMB comes from the formation of the universe.

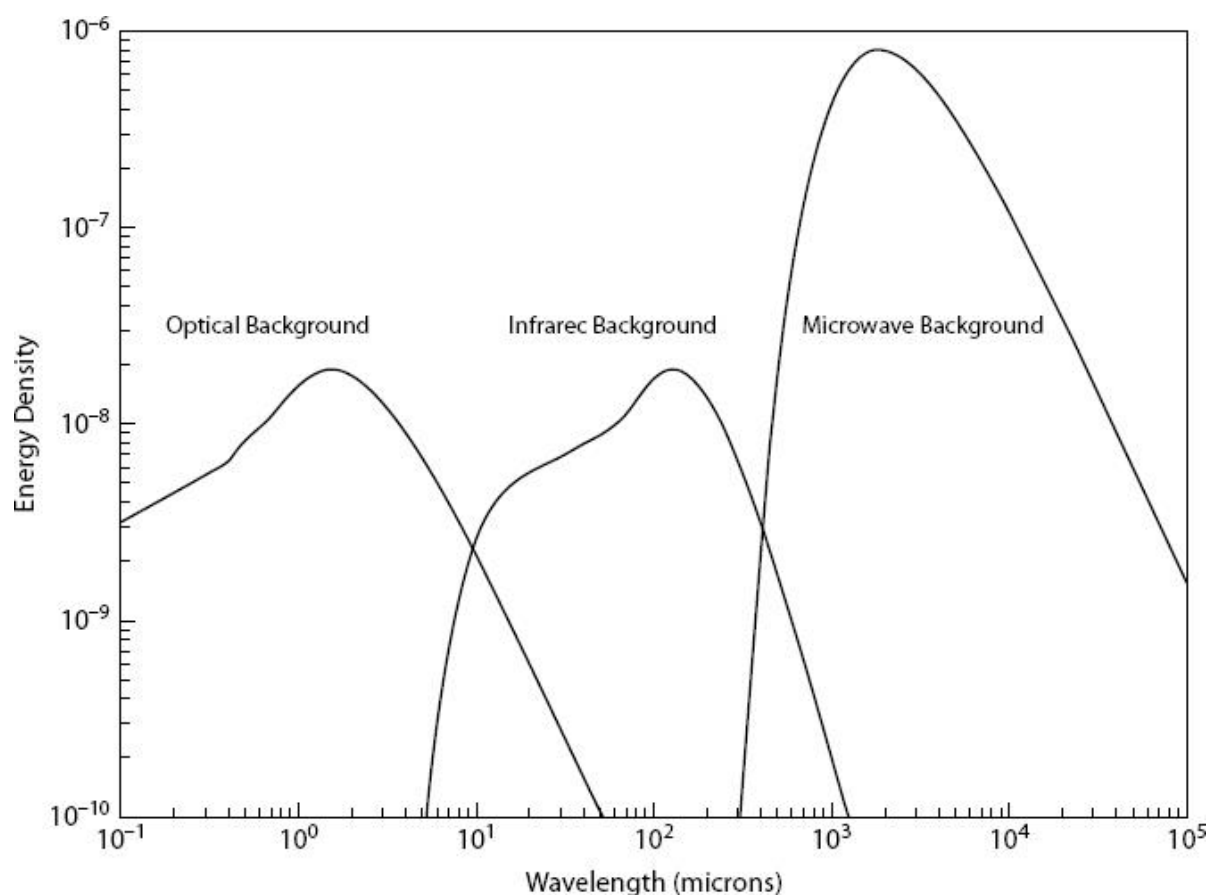


Figure 7.7 The cosmic optical, infrared and microwave backgrounds compared.

The energy density of the CMB is the highest, but the optical (COB) and infrared (CIB) backgrounds contain similar amounts of energy. (Modified from an original figure by H. Dole et al. *The cosmic infrared background resolved by Spitzer. Contributions of mid-infrared galaxies to the far-infrared background*. A&A, 451:417429, May 2006 [46]).

What are the sources that power the CIB? At the time of its discovery this was unclear, since there were no far-infrared instruments that were sensitive enough to probe the details of its structure. The suspicion, though, was that the CIB would, like the COB, be made up from contributions of many individual galaxies emitting at the appropriate wavelengths. Far-infrared light, as we have already seen, is the result of dust absorbing light from some source of energy, usually stars and star formation, and being heated to temperatures of about 50 K. The discovery that the CIB and COB each encompass comparable amounts of energy meant that half of the energy generation in the history of the universe is obscured by absorbing dust. Optical surveys, even the most sensitive ones available, will miss this energy, so the history of galaxy formation revealed by the Hubble Deep Fields is incomplete - it is telling only half of the story.

The immediate assumption of many astronomers was that the CIB and COB are generated by the same types of galaxies, and that everything they had discovered in the optical could account for the CIB just by doubling all the numbers, on the basis that half of the emission of their sources was absorbed by dust. When the first results from the new generation of instruments on ground-based telescopes capable of saying something about the sources responsible for the CIB were announced, something rather different was found.

7.7 SCUBA AND SUBMILLIMETRE GALAXIES

In the late 1990s a brand new, and much anticipated, instrument arrived at the James Clerk Maxwell telescope in Hawaii. It was called SCUBA - the Submillimetre Common User Bolometer Array [77] - and promised to revolutionise our knowledge of dusty objects. This is because it operated in the submillimetre - at wavelengths just shorter than 1 mm, where the atmosphere is reasonably transparent, but where the long wavelength tail of far-infrared emission from dust is still detectable - and was several hundred times more powerful than previous instruments at these wavelengths. It managed this huge stride in capability in two ways. Firstly, by having multiple bolometer detectors (see [Chapter 2](#) for a description of how bolometers work) - 128 in total - compared to the single detector in the previous submillimetre instrument at JCMT known as UKT-14. Secondly, each of the bolometer detectors inside SCUBA was several times as sensitive as the single detector in UKT-14. Multiply those factors together, and the stage was set for something rather special.

The Hubble Deep Field was an obvious first target for deep extragalactic observations with SCUBA. In early 1998, during the first months of SCUBA operation, a total of fifty hours of observing time was spent staring at the Hubble Deep Field [81], in what amounted to one of the single longest exposures in the nearly 9 year operational life of the instrument. The results of these observations can be seen in [Figure 7.8](#), which can be compared to the optical view of this same field in [Figure 7.1](#).

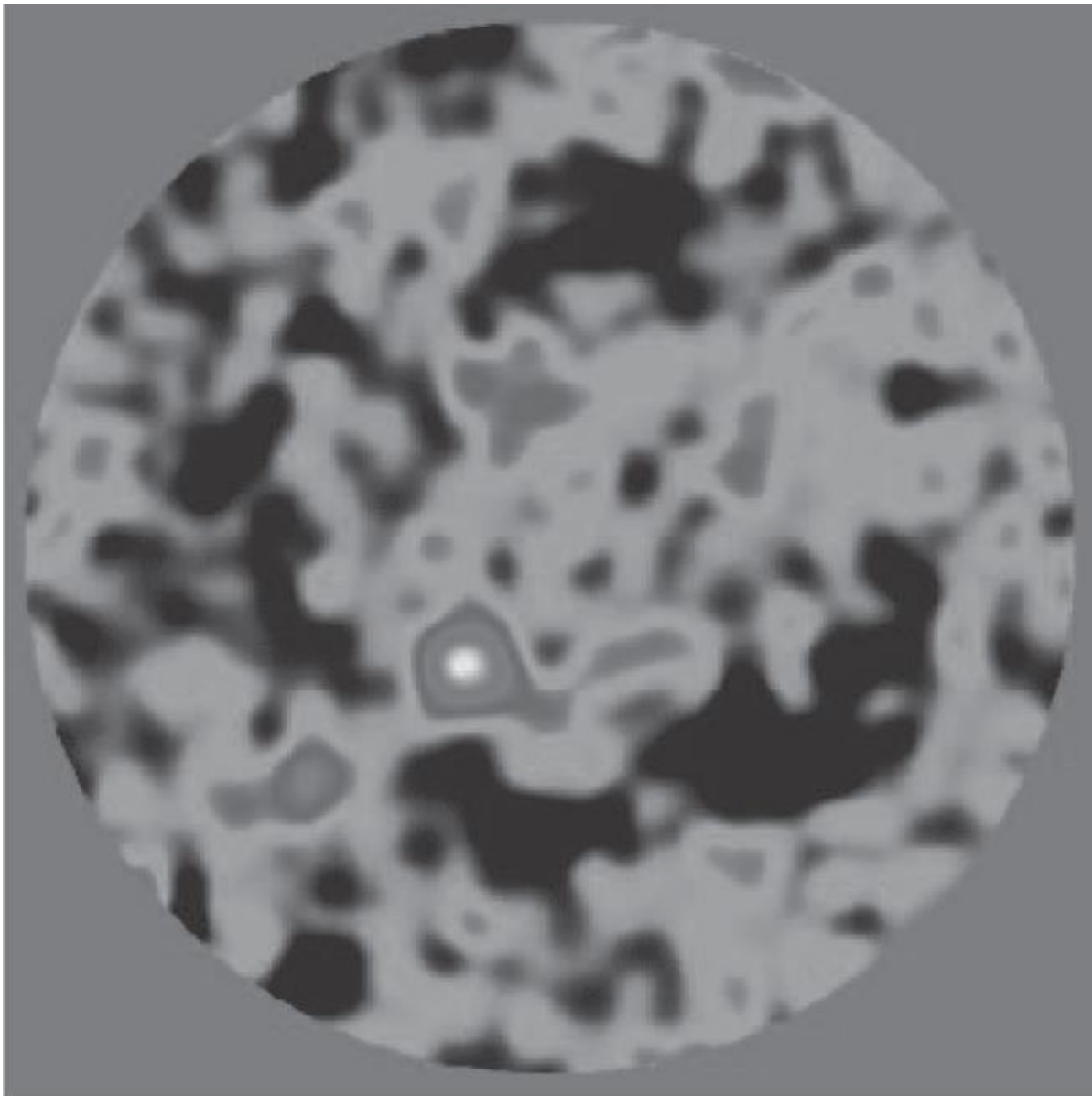


Figure 7.8 The SCUBA image of the Hubble Deep Field at a wavelength of 850 microns.

HDF850.1 is the brightest source in this image, visible as the white blob just to the lower left of the centre of the image. (Taken from [77], courtesy of the Royal Observatory Edinburgh.)

These two views of the universe, one looking at the sources that make up the COB, the other looking at the sources that make up the CIB, are very different, as you can see. While the optical image of this field reveals about two thousand relatively faint galaxies, the SCUBA image, showing emission at a wavelength of 850 microns, finds only a few sources - five in total - but each of these is individually quite bright. The picture of the galaxies responsible for the CIB that emerges from these observations is strikingly different to what has been found from the deep Hubble observations of the galaxies that make up the optical background. While the latter evolve slowly and quietly, forming stars at a moderate rate of one or two solar masses per year, the CIB galaxies, as revealed by SCUBA, are individually much more luminous. They are forming stars a hundred times faster than the galaxies behind the optical background. If we compare these distant galaxy populations to the kind of galaxies we see locally, the galaxies contributing to the optical background are like our own Milky Way Galaxy, while the CIB sources revealed by SCUBA are similar to Arp220 and other starbursts. Like the local starbursts, the phase of rapid star formation we are seeing in the SCUBA objects cannot last very long as they will use up all the gas available to them.

After the observations of the HDF with SCUBA, a new industry of surveys at submillimetre wavelengths got underway. Over the lifetime of SCUBA, the first five submillimetre galaxies (SMGs) detected in the HDF were joined by a few hundred others that were uncovered in surveys that, when they are all added together, amounted to a bit less than one square degree on the sky (the whole sky amounts to a little over 40000 square degrees). The most ambitious of these surveys, known as SHADES, short for the SCUBA Half A Degree Extragalactic Survey [114], aimed to cover half a square degree, reaching sensitivities not quite as good as the SCUBA HDF observations. These observations took place over a seven year period [36] and, as a result of technical failures that led to the retirement of SCUBA in late 2005, the full half degree area was never completed. Nevertheless, SHADES discovered over a hundred new SMGs which was, at the time, a significant step forward from the handfulls of sources found in surveys like the HDF and the Canada-UK Deep Submillimetre Survey (CUDSS) [50].

Analysis of these surveys confirmed that the population of far-infrared luminous galaxies evolved very rapidly with redshift, with this type of source being up to a hundred times more common at a redshift of two, about ten billion years ago, than they are in the local universe. The cause of this rapid evolution, and the origin of the spectacular starbursts that power both these individual objects and the CIB as a whole, is still unclear. One possibility is that galaxy interactions and mergers were much more common at earlier stages of the universe, as the galaxies we see today assembled themselves through gravitational attraction from the smaller galaxies that formed earlier in the universe. Another is that these galaxies are growing through the gravitational accretion of cool hydrogen gas from their surroundings, and that this infalling gas is fuelling bursts of star formation.

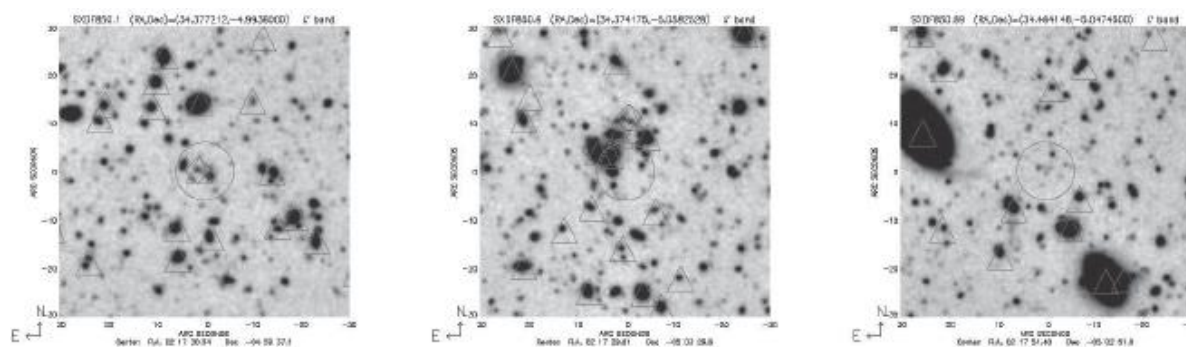


Figure 7.9 The identification of SMGs from the SHADES survey.

These show optical images (greyscale), the positions from SCUBA (circle), the positions of sources detected in the near- and mid-infrared (triangles) and radio (diamonds) for three SHADES sources. In the leftmost case a clear identification is indicated by the radio data. For the middle source there are three plausible identifications in the radio, while on the right, there is no clear identification inside the SCUBA error circle [34].

7.8 SUBMILLIMETRE GALAXIES AT OTHER WAVELENGTHS

To test the different ideas for the origin of the vast bursts of star formation that power SMGs we have to study them with the full range of observational facilities available to astronomers. Before that can be done, we have to identify their counterparts at other wavelengths. This turns out to be harder than you might think. The precision with which the position of sources can be measured with SCUBA isn't particularly good. On the basis of these positions alone, there might be several tens of possible optical counterparts for any given source in the SCUBA HDF, or in any of the other SCUBA surveys. To home in on the correct identification, we have to rely on the fact that sources that are bright in the submillimetre and far-infrared, under certain assumptions, are likely to be bright at other wavelengths, including the mid-

infrared and the radio. We can then use observations at these other wavelengths to measure the position of an SMG more precisely, and so track down their optical counterparts. An example of this process is shown in [Figure 7.9](#). These are optical images centred on the positions of sources from the SHADES survey [\[34\]](#). The positional uncertainty from SCUBA is given by the circle at the centre of the image. If you look at the leftmost part of the figure, you can see there are plenty of optical sources that lie inside this circle, and so plenty of plausible optical identifications. But there is no way of telling which of these is the correct identification using the SCUBA position and the optical data alone. The triangles indicate the positions of sources found in the SWIRE near- and mid-infrared survey using the Spitzer space telescope [\[101\]](#), while the diamonds indicate the positions of sources detected in the radio using the Very Large Array [\[86\]](#). With this extra data you can clearly see that the correct identification for the leftmost SMG is the faint optical galaxy indicated by the diamond.

Once an optical identification has been obtained, a wide range of followup observations become possible. The most important of these is the ability to measure a redshift using optical spectroscopy. In 2005, after a heroic effort at cross identification and followup optical spectroscopy, the astronomer Scott Chapman and collaborators [\[29\]](#) showed that SMGs identified in this way lay at redshifts of around 2.3. Once the distance of these sources were known, many of their physical properties, such as their size and power output, could be calculated.

However, the identification process used by Chapman and others has its flaws. As you can see in [Figure 7.9](#), while this process works well in many cases, in some, such as the source in the middle, the results can be ambiguous, while in others, such as the source on the right, the process fails completely. At redshifts above 3, the brightness of the sources in the radio or near- and mid-infrared are expected to fall off. More distant sources, which will also be the most powerful and most interesting, are likely to be missed by this particular approach.

7.9 THE HDF850.1 STORY

The brightest submillimetre source found in the Hubble Deep Field, given the prosaic name of HDF850.1, is a great illustration of the problems of identifying the most distant submillimetre galaxies. This source was originally found in the HDF SCUBA observations in 1998 [\[81\]](#). As the brightest source found, it might be expected to be the brightest at other wavelengths as well, but this turned out not to be the case. Over subsequent years there were observations at a range of different observatories and wavelengths. A better position was obtained from millimetre interferometers [\[49\]](#), which led to claims that it was associated with an optical source at a redshift of about 1.7, but there remained a significant disagreement between the millimetre and optical position. More and more sensitive observations at a range of wavelengths were acquired, from X-ray to radio, but still no convincing identification was made. There were even suggestions that HDF850.1, and, by implication, many of the other unidentified SMGs, weren't galaxies at all, but were cold dusty gas clouds in our own galaxy [\[96\]](#).

Eventually, a campaign of millimetre spectroscopic observations, scanning a wide range of frequencies for an emission line from the carbon dioxide molecule, made a detection [\[171\]](#) and allowed a redshift to be measured, as well as a much more accurate position. The redshift of 5.2, meaning that this object is seen just a billion years after the Big Bang, was surprisingly high. To be as bright as it was seen by SCUBA at such a high redshift, HDF850.1 must be forming stars at a rate of almost 1000 solar masses of new stars each year - nearly a thousand times faster than the rate of star formation in the Milky Way or in a typical optical/near-infrared detected galaxy from the Hubble Deep Fields. Just as surprising was the fact that no optical or near-infrared emission could be seen at the position of HDF850.1, even in the deepest Hubble Space Telescope data available. This galaxy must have so much gas and dust in its interstellar medium that most of its starlight is absorbed by dust and re-emitted in the far-infrared.

The fact that it took fourteen years to get a redshift measurement of HDF850.1 is a salutary warning for all far-infrared and submillimetre astronomers. Sometimes the universe doesn't put your sources where you think they should be, and doesn't make them easily detectable at other wavelengths. HDF850.1 was one of the first five such objects to be found. All the SCUBA surveys combined have found a few hundred more, and other similar instruments that have come along since then have added a few hundred more. Chasing all of those down the way HDF850.1 was eventually dealt with could take a long time indeed. However, catalogs of a few hundred SMGs can now be considered pitifully small. With the Herschel Space Observatory we can now find *hundreds of thousands* of these objects.

7.10 THE HERSCHEL REVOLUTION

The Herschel Space Observatory, shown in [Figure 7.10](#), was launched on 14th May 2009. It was the culmination of many decades of planning - the original ideas for a space based far-infrared telescope date back to the early 1980s, and, once fully approved, it took about ten years to build and get ready for launch. Herschel is one of the largest space astronomy missions ever undertaken by the European Space Agency. Its most obvious unique feature is a huge, 3.5 m diameter, primary mirror, the largest mirror ever launched on an astronomical telescope³. Previous infrared astronomy satellites all had much smaller mirrors, with the largest of those being Spitzer at 85 cm. The large mirror allows Herschel to collect nearly seventeen times as much light as Spitzer. When this is combined with more modern detectors, and instruments that extend to longer wavelengths, major advances are possible. You will have seen some of these results in previous chapters but, to my mind at least, it is in the observations of SMGs where Herschel's capabilities are shown most clearly.

Of the three scientific instruments on Herschel, the most important for detecting large numbers of SMGs is SPIRE - the Spectral and Photometric Imaging Receiver [\[70\]](#). This instrument, developed largely in the UK, uses bolometers to image the sky simultaneously at three separate far-infrared/submm wavelengths - 250, 350 and 500 microns. The combination of Herschel's large mirror, the absence of atmospheric absorption and lack of a large thermal background mean that SPIRE can scan large areas of the sky very quickly. Over its entire career, SPIRE images were taken which covered almost a tenth of the entire sky. The two most important projects in the context of searching for distant SMGs are the HerMES [\[125\]](#) (Herschel Multitier Extragalactic Survey) and H-ATLAS (Herschel Astrophysical Terahertz Large Area Survey)⁴ surveys [\[51\]](#). These take two different but complementary approaches to studying distant far-infrared luminous galaxies. The HerMES survey aims to be a little bit like the Hubble Deep Fields, taking sensitive observations in parts of the sky that are already well studied at other wavelengths. It covers a total of about 100 square degrees, with a more recent and less sensitive extension of a further 270 square degrees. The H-ATLAS survey, in contrast, aims for somewhat less sensitive observations but, at 570 square degrees, it covers a much larger area. Both the surveys are in areas of the sky free from contamination by the dust inside our own galaxy, making them ideal for extragalactic astronomy. Between them, HerMES and H-ATLAS have scanned nearly a thousand square degrees, roughly 5% of the sky suitable for extragalactic astronomy, and nearly a thousand times the area covered so far by ground-based surveys in the submillimetre using instruments like SCUBA.

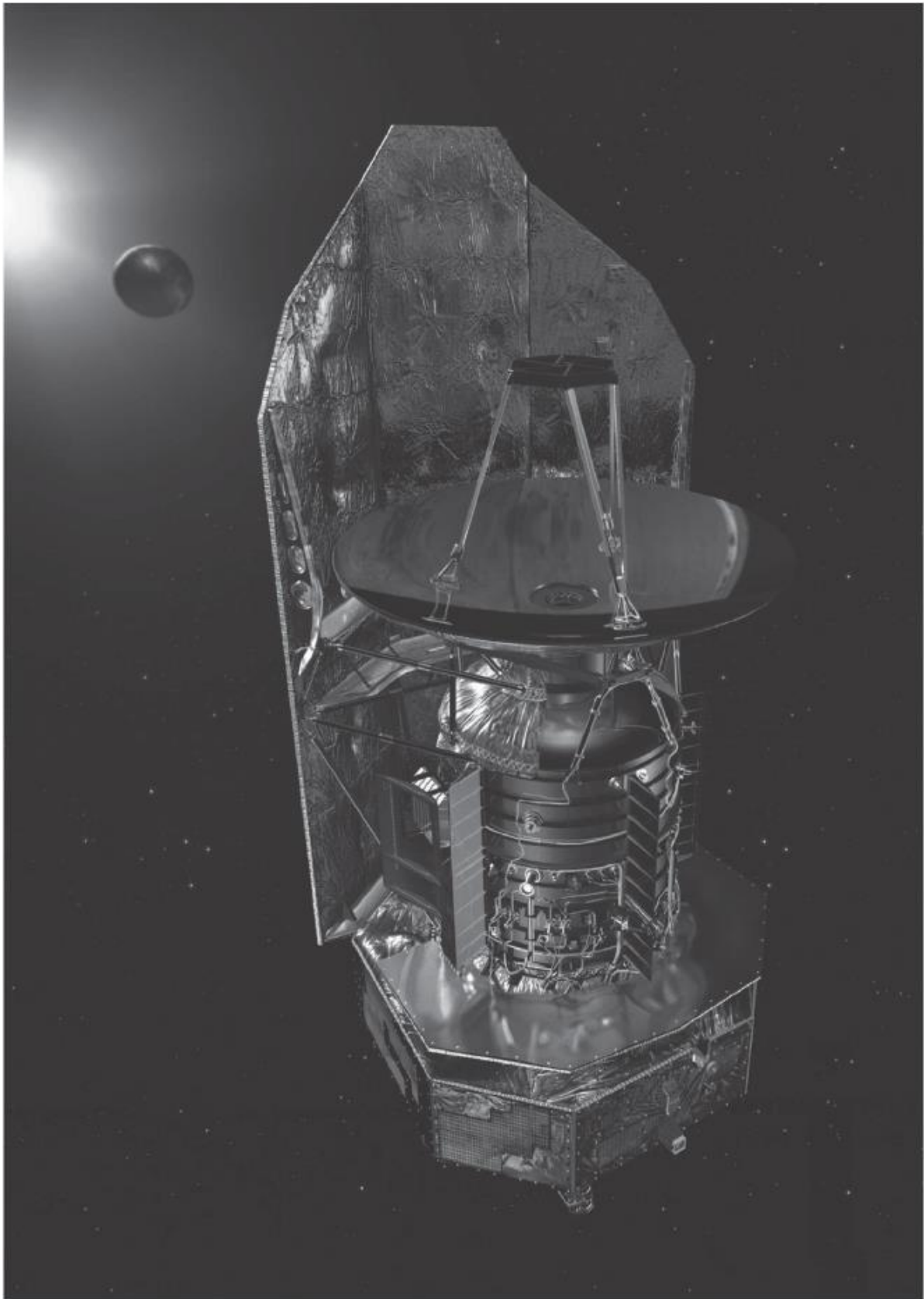


Figure 7.10 The Herschel Space Observatory.

(Courtesy of ESA <http://www.esa.int/ESA>.)

[Figure 7.2](#) gives an impression of just how much of an advance the SPIRE surveys are over what has come before. The SCUBA image of the Hubble Deep Field, shown to scale as an inset, took fifty hours of observing time, and found five galaxies. The much larger HerMES image, a small fraction of the survey as a whole, contains about 700 galaxies and took

about eight hours of observations. In total the H-ATLAS and HerMES surveys have detected several hundred thousand SMGs. The days when you could recall the details of every single known SMG are now gone. We now have to look at these objects not as individuals, but as a population, in much the same way that we can look at stars or local galaxies. This really is a revolution, and the final results of this vast new database are still to be seen.

7.11 THE MOST DISTANT DUSTY GALAXIES

One of the things made possible by the huge catalogs of SMGs that are now available thanks to Herschel is that we can start looking for the most exceptional objects. The fact that SPIRE works at three different frequencies simultaneously is a great help. If we look at the typical spectral energy distribution for a star-forming galaxy at low redshift (see [Figure 6.5](#)) you will see that such a source will appear ‘blue’ in the SPIRE bands, meaning that it looks brightest at the shortest wavelengths available. At higher redshifts, though, this colour will change, as shown in [Figure 7.11](#).

A local dusty galaxy will appear brighter in the 250 micron channel than at 350 or 500 microns. At a redshift of three, though, at the upper range of redshifts measured for the SCUBA detected and radio/mid-infrared identified SMGs, it will have roughly the same brightness in the 250 and 350 micron channels. At still higher redshifts, beyond that of HDF850.1, the 500 micron channel will be the brightest. So, as the source moves to higher redshift, the peak of the dust emission, which lies at a wavelength of about 100 microns when it is emitted, will be shifted through the SPIRE bands. Any sources that appear red, in other words sources that are brightest at 500 microns, may well lie at redshifts beyond those where SMGs have previously been found.

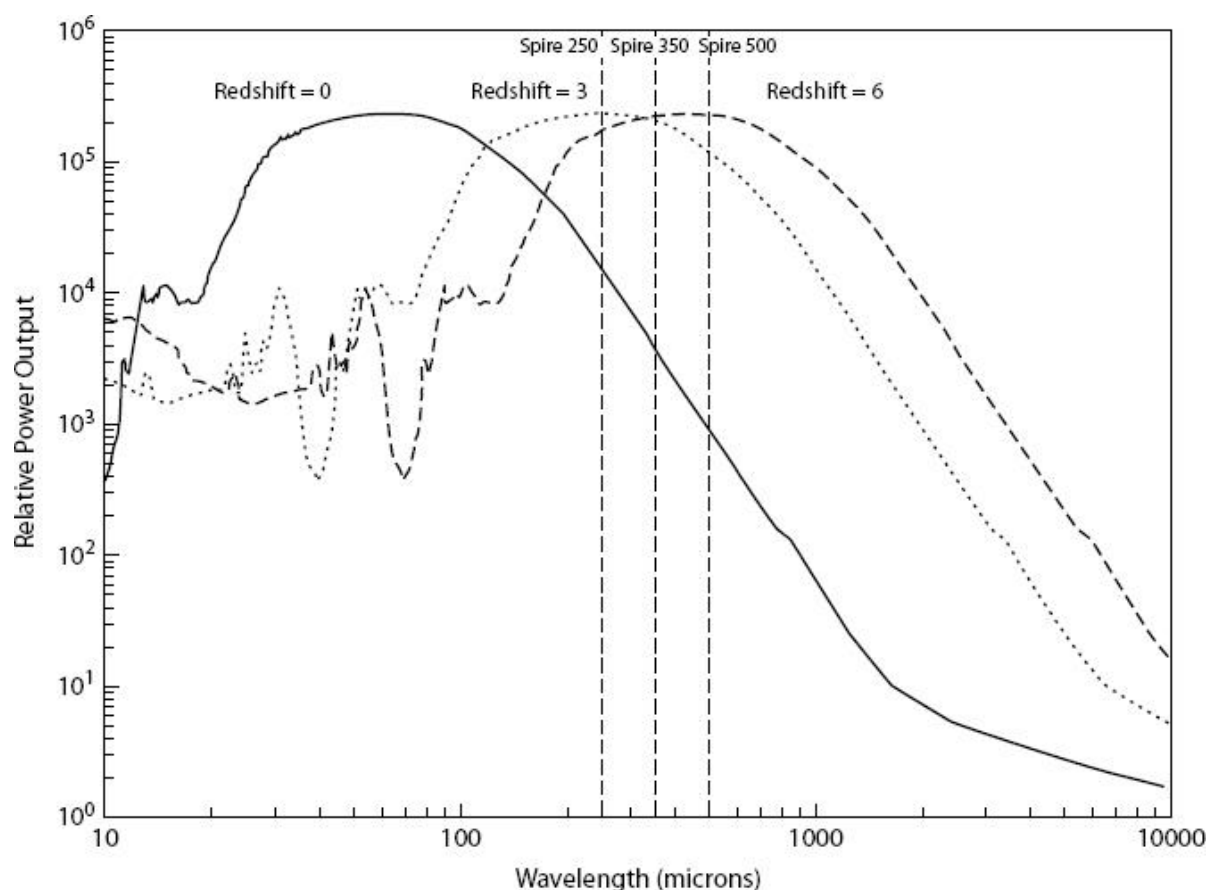


Figure 7.11 How the SED of a dusty galaxy changes with redshift.

This shows how the SED of a dusty star-forming galaxy like Arp220 changes its appearance with redshift. The SPIRE bands are shown with vertical dashed lines, and the galaxy SED is shown at three different redshifts: solid for a redshift of zero, dotted for a redshift of three, and dashed for a redshift of six.

This colour selection method is actually quite simple to apply to catalogs of SPIRE sources and to SPIRE images. The HerMES team did this to some of the first data that arrived from Herschel and were quite surprised to find more red SPIRE sources than they expected. One of these sources, given the name of HFLS3, was exceptionally red (see in [Figure 7.2](#)) and was the subject of a multi-telescope followup campaign to confirm its existence and measure its redshift, which was found to be 6.34 [[141](#)], corresponding to a time less than a billion years after the Big Bang. This made it not only the most distant dusty star-forming galaxy currently known, but also put it among the most distant galaxies for which a spectroscopically confirmed redshift is available. Among the many surprising things about HFLS3 is the rate at which it is forming stars - about 2000 solar masses of new stars per year. On its own, HFLS3 is forming stars at a rate comparable to that of all galaxies at a similar redshift that were found in the optical/near-infrared images of the Hubble Deep Field. The mass of dust in this galaxy, about a billion solar masses, is also a problem. The carbon and silicon that makes up this dust has to have been made since the Big Bang, fused from primordial hydrogen and helium in stars. These stars have less than a billion years from the Big Bang until the time at which we see HFLS3 to form, live out their lives, and recycle their enriched material back into the interstellar medium as dust. Clearly stars like our own Sun, which has an expected lifespan of about 10 billion years, are not able to do this in time, so we are beginning to learn something about the very first generation of stars to form in the universe, so called Population 3 stars, even though we have yet to detect any of them directly.

Work is still underway to fully understand the other red sources seen in the HerMES studies which found HFLS3. The search for red Herschel sources is also being extended across the full HerMES and H-ATLAS surveys. A number of other galaxies at redshifts greater than four have already been confirmed, though HFLS3 remains the record holder. Taking these measured redshifts as confirmation that most of these sources are genuinely at redshifts greater than four, we can start to define the properties of this population of dusty, rapidly star-forming galaxies. We can also compare the number of sources we see to the number predicted by the best existing models of galaxy formation. It turns out that none of these models predict the number or brightness of red sources that the Herschel surveys are finding [[48](#)]. This presents an interesting problem for those modelling galaxy formation. We can also look at the contribution of these galaxies, which are rare but which are forming stars very rapidly, to the star-formation history of the universe. At first sight, the high redshift Herschel sources only contribute about 10% of the star-formation rate of optical/near-infrared sources at a similar redshift. However, the sources that Herschel can detect in this way have to be very luminous. If you assume that there is an underlying population of similar but fainter objects that Herschel cannot directly detect, you find that these high-redshift dusty star-forming galaxies might contribute as much star formation as the sources found at similar redshifts through optical/near-infrared observations. Once again, it looks as if the optical is only telling us half the story, and our picture of the history of star formation in the universe will only be complete when we also understand what is happening in the far-infrared.

7.12 GALAXIES AS TELESCOPES

While Herschel allows us to detect very distant, very powerful far-infrared and submillimetre galaxies, it is not capable of providing the kind of detailed, high resolution view of these sources that Hubble can provide for the sources it detects. The Hubble Space Telescope has an angular resolution that can be measured in the ten millionths of a degree. Herschel can only resolve structures a hundred times bigger. Also, as we have seen in the previous section, Herschel can only detect the most powerful dusty galaxies at high redshifts, sources like HFLS3 which is forming stars at a massive 3000 solar masses of new stars each year. To be able to understand the processes that drive these sources, to see if their bursts of star formation are triggered by galaxy interactions and mergers, as is the case for their lower redshift, lower luminosity equivalents like Arp220, we need to be able to examine them with much greater precision. New instruments, such as the Atacama Large Millimetre Array (ALMA) are currently being constructed that will be able to do this, but, in the interim, there is another way for us to get a close up view of these sources. This approach also has the potential to tell us something about the mysterious dark matter which makes up much of the mass of the universe.

One of the key predictions of Einstein's General Theory of Relativity is that gravity bends the path of light rays in a process known as gravitational lensing. The degree to which mass bends light is actually quite small, so this effect is only visible when light passes close to a massive object. This can be seen when light passes close to a star like the Sun, which slightly distorts the view of stars close behind it, but it is seen most clearly when light from a distant galaxy passes close to another galaxy or through a cluster of galaxies. An example of the latter can be seen in [Figure 7.14](#), where light from background galaxies passes through the galaxy cluster RCS2 032727-132623, producing a variety of spectacular arcs. These are the distorted images of the lensed background galaxies. Gravitational lensing not only makes the background galaxies brighter, it also spreads their light out over a much larger area. This allows much smaller structures in the lensed background galaxy to be seen than would otherwise be possible. Since the strength of lensing depends on the gravitational mass of material bending the light, you can also use lensing images, like [Figure 7.14](#), to measure the distribution of mass in a galaxy or galaxy cluster. Since most of this mass is made up of dark matter, not the ordinary matter that you, me, the Earth and Sun are made of, determining the distribution of mass inside objects like galaxies and galaxy clusters is a first step to working out what the dark matter actually is, and how it behaves.

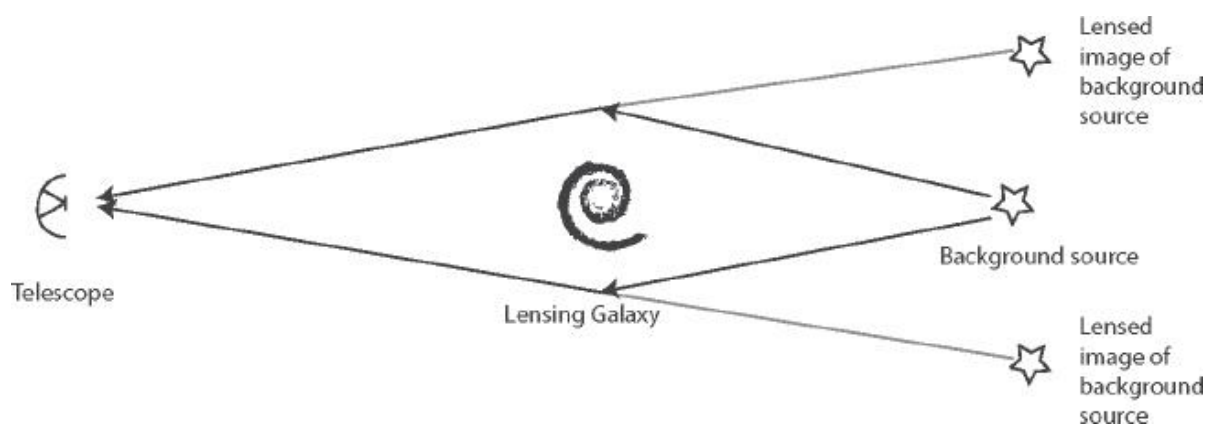


Figure 7.12 Strong gravitational lensing.

Diagram showing how galaxy-galaxy lensing produces multiple images of a background source.

Galaxy clusters can strongly magnify a number of objects behind them, as you can see in [Figure 7.14](#), but they are relatively rare objects. Individual galaxies, which are much more common, can also provide lensing magnification when a distant object lies right behind them. This is known as galaxy-galaxy lensing, and a diagram showing how it functions can be seen in [Figure 7.12](#). A source lying behind a foreground lensing galaxy can, in this way, appear as multiple images. If the geometry is exactly right, the light from the background source is spread out into a ring that surrounds the lensing

galaxy. This ring is known as an Einstein ring. As with galaxy cluster lensing, the background source is magnified so it appears brighter than it would otherwise be, and its light is spread out so that smaller scale features can be seen. You can also use the shape of the lensed source to work out the amount and distribution of mass in the lensing galaxy. This means that galaxy-galaxy lensing is potentially a very powerful tool as it can provide a lot of extra information about both the background lensed source, and about the dark matter in the lensing galaxy. If a large number of galaxy-galaxy lenses were available, with the lensing galaxies at a range of redshifts, you could, for example, use them to see how the mass of galaxies grows with time, irrespective of whether that mass is luminous, in the form of stars, or dark matter.

While studies of galaxy-galaxy lenses are potentially very powerful, they are not very easy to find. The efficiency of previous lensing searches, in terms of the number of lenses found compared to the number of candidates studied, has been pretty low. The CLASS lensing survey (Cosmic Lens All Sky Survey), which used radio data, found 22 lensed systems in a survey of 16,000 candidates [116], giving an efficiency a bit better than one in a thousand. Optical searches for strong lenses using the Sloan Digital Sky Survey (SDSS) have found 62 lenses from a sample of about 51,000 candidates [84] for a similar efficiency. Studying the images of potential lensing galaxies in deep Hubble imaging fields has produced twenty lenses from about 9000 candidates, for an efficiency of about one in 500 [56]. The most efficient lens survey to date is probably the Sloan Lens ACS survey (SLACS), which uses a combination of optical spectroscopy from the SDSS and imaging from the Hubble Space Telescope [9]. This has found about 100 lenses with an efficiency of about 68% but relies on both the large existing database of the SDSS, and on expensive HST followup observations. These searches have found a total of about 200 strong galaxy-galaxy lenses [163] which have begun to reveal interesting results about the distribution of mass in the lenses, and about the background lensed sources. What is needed to significantly advance the field is a larger and cleaner sample of galaxy-galaxy gravitational lensing systems, and the central obstacle to this is the low efficiency of most selection methods.

When Herschel was launched, not many people thought it would prove to be useful for gravitational lens searches. However, with hindsight, its potential for pinpointing lenses should have been clear. The key to this is the fact that the background, distant dusty galaxies, discussed in the previous section, are fairly numerous, are quite faint, and have a distinctive colour. Because of this, exceptionally bright sources with the same distinctive red colour are unlikely to be intrinsically bright - sources like this just don't exist. Instead, they appear to be bright because their emission is being magnified by a foreground galaxy.

The first data that arrived from the H-ATLAS survey was searched for candidate gravitational lenses. The H-ATLAS team selected the brightest eleven objects from the 7000 sources detected in the first 15 square degrees of the 570 square degree survey. One of these sources turned out to lie within our own galaxy, one was an unusual radio source (a blazar, see [Chapter 6](#)) and four were easily identified as nearby galaxies. That left five remaining candidates. These were all found to have the red colours characteristic of high redshift dusty galaxies. Followup observations of these sources [118] found that every single one of them was the result of strong gravitational lensing, meaning that Herschel has provided us with a near 100% efficient way of finding gravitational lenses. Further observations of these candidates have found some spectacular examples of multiple images and one Einstein Ring - see [Figure 7.16](#).

The exploitation of this technique is still in its early days, but it seems very likely that a large number of lensed systems will be found, stretching out to high redshift for both the lens and the background source. This will be ideal for understanding the assembly of mass over time, and in examining the details of the distant dusty galaxies magnified by gravitational lensing.

7.13 ALMA THE NEXT REVOLUTION

Herschel isn't the only far-infrared/submm telescope that has proved to be ideal for finding gravitational lenses. The South Pole Telescope (SPT), an instrument designed mainly to study the CMB, uncovered a number of unusually bright point sources whose colours suggested they might be gravitational lenses. Followup observations with the Atacama Large Millimetre Array (ALMA) showed that these sources were all gravitationally lensed [169]. Even though ALMA was only a quarter completed at the time of the observations, it was still able to produce some of the best images of lensed dusty systems ever seen. Some of these are shown in [Figure 7.15](#).



Figure 7.13 Some of the many dishes of ALMA.

(Courtesy of ALMA <http://www.almaobservatory.org/>.)

ALMA will be the next great far-infrared/submillimetre telescope, and the observations of the SPT lenses are just the first glimpse of its capabilities. ALMA will be made up of 66 antennas spread across a plateau 5000 m up in the Atacama desert of Chile (see [Figure 7.13](#)). The 66 dishes give it a much greater collecting area than any previous millimetre or submillimetre telescope, and the high, dry location will minimise the effects of atmospheric absorption. ALMA is an interferometer, which means that the light from separate antennas is combined to provide an angular resolution equivalent to that of a single dish that matches the separation of the antennae. Since ALMA's antennae can be up to 16 kilometres apart, this means that finally we will get resolutions in the far-infrared and submillimetre that match, or even exceed, those that optical and near infrared astronomers can achieve with the Hubble Space Telescope.

7.14 CONCLUSIONS

The story of galaxy formation and galaxy evolution is not complete. As we have seen, we now know that our traditional optical and near-infrared observations, with large ground-based telescopes or the Hubble Space Telescope, tell only half of the story, and we need the far-infrared and submillimetre to tell us what happens in the dusty objects responsible for producing the CIB. Only with Herschel has it been possible to find the large numbers of far-infrared sources necessary to really understand what is going on, but our work with the Herschel data is far from complete.

The Herschel space telescope stopped operation at the end of April 2013, after nearly four years in space. It has left a huge legacy of observations, much of which, like the HerMES and H-ATLAS projects, is concerned with the high redshift universe. Meanwhile, new ground-based telescopes such as ALMA, and next-generation instruments such as SCUBA2, the successor to the immensely successful SCUBA, are now coming on stream.

While the story of the dusty, high redshift universe is not yet complete, we now have the tools to fully get to grips with the history of the obscured universe.



Figure 7.14 Gravitational lensing in the galaxy cluster RCS2 032727-132623.

This shows a Hubble Space Telescope optical image of the galaxy cluster RCS2 032727-132623. Galaxies behind this cluster are gravitationally lensed by its large mass, producing the spectacular arc-like structure in the image. (Courtesy of NASA.)



Figure 7.15 ALMA images of the SPT lenses.

The dust emission from the lensed sources is shown here in red, while blue shows HST optical emission from foreground

sources and the lenses themselves. As you can see, the lensed objects appear as multiple images, rings and arcs, exactly what you expect to find with galaxy-galaxy gravitational lensing. (Courtesy of ALMA <http://www.almaobservatory.org/> [[169](#)].)



Figure 7.16 Hubble Space Telescope image of one of the first H-ATLAS lenses.

Light from the lensed source forms the ring around the elliptical galaxy at the centre of the image. (Image used by kind permission of Mattia Negrello and the H-ATLAS Consortium.)

¹ It wasn't one of my best.

² So closely, it turned out, that the CMB was a better black body than the calibrator on the instrument.

³ It might be the largest mirror ever launched into orbit, but there are persistent rumours that various spy satellites, looking down rather than up, have still larger mirrors. But nobody is giving details of these, and they're certainly not making them available for astronomers to use.

⁴ Any acronym that uses the word Terahertz is clearly just a little bit too contrived.